CYCLIC ORDERINGS OF LAMINAR MATROIDS WITH PRESCRIBED BASIS INTERVALS

KNOWLEDGE SENSE

Abstract. Let M be a laminar matroid of rank r whose ground set is partitioned into bases $B_1, \dots, B_k$. We show that the elements of each B_i can be ordered so that, in the cyclic sequence obtained by listing $B_1, \dots, B_k$ in this order, every r cyclically consecutive elements form a basis of M. Since the indexing of the bases is arbitrary, the blocks may appear in any prescribed cyclic order. This answers, for laminar matroids, a question of Wiedemann. It also proves, for this class, a conjecture of Bérczi, Jánosik and Mátravölgyi that each block can be made an interval, together with the prescribed block order that they obtained for split matroids. The proof is an induction over the tree of a laminar presentation, in which the orders obtained for the children of a node are merged by a persistence rule. The construction is explicit and can be carried out in polynomial time.

1. Introduction

Let M be a matroid of rank r on a finite set E. A *cyclic ordering* of M is a cyclic sequence containing every element of E exactly once, in which every r cyclically consecutive elements form a basis. If such an ordering exists and r divides $|E|$, then cutting the cycle into consecutive pieces of length r partitions E into bases. The matroid M is *uniformly dense* if $r \cdot |X| \le |E| \cdot r(X)$ for every $X \subseteq E$. Every matroid with a cyclic ordering is uniformly dense, and Kajitani, Ueno and Miyano [8] conjectured the converse; van den Heuvel and Thomassé [7] proved it when $|E|$ and r are coprime. For a partition $E = B_1 \cup B_2$ into two bases, Gabow [6] conjectured that there is a cyclic ordering in which B_1 and B_2 are intervals; see [1, Conjecture 1].

The case of many bases goes back to a manuscript of Wiedemann [12]. There he shows that for any bases $B_1, \dots, B_\ell$ of a matroid and any order of B_1, the remaining bases can be ordered so that every r consecutive elements of the *linear* concatenation $B_1 B_2 \cdots B_\ell$ form a basis [12, Section 2]. Only the passage from B_ℓ back to B_1 is missing, and Wiedemann writes that it may be possible to produce a cyclic ordering from *any* cyclic ordering of a partition into bases. He also proves that a prescribed partition $B_1 = X_1 \cup Y_1$ extends to partitions $B_i = X_i \cup Y_i$ such that every $Y_i \cup X_{i+1}$ is a basis, cyclically [12, Section 3]; this concerns one cut of each block rather than all cuts simultaneously. Bérczi, Jánosik and Mátravölgyi [1, Conjecture 3] conjectured that for every partition $E = B_1 \cup \cdots \cup B_k$ into bases there is a cyclic ordering in which each B_i is an interval. They proved this for split matroids, and their proof keeps the blocks in the order $B_1, \dots, B_k$ [1, Theorem 4 and Remark 5]. Our main result proves this conjecture for laminar matroids, with the blocks in any prescribed cyclic order, as in Wiedemann's question.

Theorem 1.1. *Let M be a laminar matroid of rank r on a finite set E, and let $B_1, \dots, B_k$, $k \ge 1$, be bases of M that partition E. Then each B_i has a linear order such that, in the cyclic*

Date: 8 October 2026.

2020 *Mathematics Subject Classification.* Primary 05B35; Secondary 05A05.

Key words and phrases. Matroid, laminar matroid, cyclic ordering, basis partition, Gabow's conjecture, strongly base orderable matroid.

sequence obtained by concatenating $B_1, B_2, \dots, B_k$ in this order, every r cyclically consecutive elements form a basis of M.

Since the indexing of the bases is arbitrary, the blocks can be made to appear in any prescribed cyclic order. Only the orders inside the blocks are chosen. There is no restriction on r, on k, or on the shape of a laminar presentation of M. In particular, every laminar matroid whose ground set is a disjoint union of bases has a cyclic ordering, and laminar matroids satisfy [1, Conjecture 3].

Known cases. For $k \leq 2$, Theorem 1.1 follows from strong base orderability (Section 5.1): laminar matroids are gammoids [4], see [3, Section 1], and gammoids are strongly base orderable, see [2, Section 1]. For transversal matroids, a short matching argument gives the analogue of Theorem 1.1 for all k (Section 5.2). Neither argument covers laminar matroids with $k \geq 3$. The rank-three laminar matroid with twelve elements obtained as the case $k = 4$ of a construction of Bérczi, Schwarcz and Yamaguchi [2, proof of Theorem 1.10] is neither transversal nor split, as we show in Proposition 5.2; so Theorem 1.1 is not contained in the transversal case or in the split case of [1]. We are not aware of a previous proof of Theorem 1.1 for laminar matroids with $k \geq 3$.

Idea of the proof. A laminar matroid is described by a tree of capacity constraints (Section 2). We order the elements within each block by induction over this tree, from the leaves to the root; the order of the blocks themselves stays as prescribed. Restricted to a child of a node, the blocks are independent sets of different sizes rather than bases, so the induction is carried out for a stronger statement about arbitrary disjoint independent sets (Proposition 4.1). At a node, the orders already obtained for the children are merged: we label the j-th element of a child u in every block by (u, j), so that the same label occurs in many blocks, and in each block we sort the labels by the number of following blocks in which they persist. The key property (Lemma 3.1) is that a label occurring in two consecutive blocks never occupies an earlier position in the second block. This monotonicity reduces every set of consecutive elements of the merged sequence to such sets for the children.

Organization. Section 2 introduces capacity trees and the windows of a cyclic list of rows. Section 3 proves the two lemmas on the persistence order. Section 4 proves the main proposition and Theorem 1.1. Section 5 discusses the known cases, the separating example, and further remarks.

2. Capacity trees and windows

Our matroid terminology follows Oxley [10]. Let $\mathcal{A}$ be a laminar family of subsets of a finite set E, that is, any two members of $\mathcal{A}$ are disjoint or one contains the other, and let $c\colon \mathcal{A} \to \mathbb{Z}_{\geq 0}$. The sets $I \subseteq E$ with $|I \cap A| \leq c(A)$ for all $A \in \mathcal{A}$ are the independent sets of a matroid $M(E, \mathcal{A}, c)$, and a matroid is *laminar* if it is isomorphic to such a matroid [3, Section 1]. It is convenient to organize the constraints as a tree.

Definition 2.1. A *capacity tree* on a finite set E is a finite rooted tree in which every node v carries a set $E_v \subseteq E$ and an integer $c_v \geq 0$, such that $E_v = E$ for the root v and, for every node v that is not a leaf, the sets E_u of the children u of v partition E_v. Distinct nodes may carry the same set. For each node v we define a family $\mathcal{I}_v$ of subsets of E_v recursively: if v is a leaf, then $X \subseteq E_v$ belongs to $\mathcal{I}_v$ if $|X| \leq c_v$; otherwise $X \subseteq E_v$ belongs to $\mathcal{I}_v$ if $|X| \leq c_v$ and $X \cap E_u \in \mathcal{I}_u$ for every child u of v.

By induction from the leaves, a set $X \subseteq E_v$ belongs to $\mathcal{I}_v$ if and only if

$$|X \cap E_w| \le c_w \quad \text{for every node } w \text{ of the subtree rooted at } v. \tag{2.1}$$

The sets E_w of this subtree form a laminar family. If a set occurs for several nodes, only the smallest of their capacities matters in (2.1). Hence $\mathcal{I}_v$ is the family of independent sets of the laminar matroid on E_v defined by these sets, each with this smallest capacity. We denote this matroid by N_v and call the members of $\mathcal{I}_v$ independent in N_v. Every independent set of N_v has at most c_v elements. Note that N_v need not be the restriction to E_v of the matroid N_{root}, because an ancestor of v may impose a stronger bound. For instance, if the root has capacity 1 and ground set $\{a, b, d\}$ and has the children $\{a, b\}$ of capacity 2 and $\{d\}$ of capacity 1, then $N_{\{a,b\}}$ is free, whereas a and b are parallel in N_{root}. The induction below takes place in the matroids N_v.

Lemma 2.2. *Let M be a laminar matroid of rank r on E. Then there is a capacity tree on E whose root has capacity r and satisfies $N_{\mathrm{root}} = M$.*

Proof. We may assume $M = M(E, \mathcal{A}, c)$ with nonnegative integer capacities [3, Section 2] and $\emptyset \notin \mathcal{A}$. Put $\mathcal{A}' = \mathcal{A} \cup \{E\}$, let $c'(E) = r$, and let $c' = c$ on $\mathcal{A}' \setminus \{E\}$. Every independent set of M has at most r elements, and if $E \in \mathcal{A}$ then $r \le c(E)$ because a basis satisfies the constraint for E. Hence c' defines the same independent sets as c. The family $\mathcal{A}'$ is laminar, so for $A \in \mathcal{A}' \setminus \{E\}$ the members of $\mathcal{A}'$ properly containing A form a chain; its smallest member is the parent of A. This makes $\mathcal{A}'$ a rooted tree with root E. Finally, for every $A \in \mathcal{A}'$ that has children and for which $R_A = A \setminus \bigcup\{\text{children of } A\}$ is nonempty, we add a leaf with ground set R_A and capacity $|R_A|$. The result is a capacity tree. The added leaves impose no condition, so by (2.1) the independent sets of N_{root} are those of $M(E, \mathcal{A}', c')$, which are the independent sets of M. □

This tree reflects the description of laminar matroids as the matroids built from the empty matroid by adding coloops, truncating and taking direct sums [3, Theorem 1.5].

We now fix notation for cyclic lists of linearly ordered sets. For a linear order L of a finite set I and an integer $t \ge 0$, let $L^{\le t}$ be the set of the first $\min(t, |I|)$ elements of L, and let $L^{>t} = I \setminus L^{\le t}$. The *position* of $x \in I$ in L is s if x is the s-th element of L; thus $x \in L^{\le t}$ if and only if its position is at most t.

Definition 2.3. Let $c \ge 0$ and $k \ge 1$, and let L_i be a linear order of a finite set I_i with $|I_i| \le c$, for $i \in \mathbb{Z}_k$. For $i \in \mathbb{Z}_k$ and $0 \le t \le c$ the *window* is

$$W_{i,t} = L_i^{>t} \cup L_{i+1}^{\le t}.$$

We call the linear orders L_i the *rows*; they are indexed cyclically by $\mathbb{Z}_k$.

Lemma 2.4. *In the situation of Definition 2.3, $|W_{i,t}| \le c$. If $c > 0$ and $|I_i| = c$ for every i, then the sets of c cyclically consecutive entries of the cyclic sequence $L_1 L_2 \cdots L_k$ are exactly the sets $W_{i,t}$ with $i \in \mathbb{Z}_k$ and $0 \le t \le c - 1$.*

Proof. If $t \le |I_i|$, then $|W_{i,t}| \le (|I_i| - t) + t \le c$; otherwise $L_i^{>t} = \emptyset$ and $|W_{i,t}| \le |I_{i+1}| \le c$. If all rows have length c, the c consecutive entries starting at the $(t+1)$-st entry of L_i, where $0 \le t \le c - 1$, are the last $c - t$ entries of L_i followed by the first t entries of L_{i+1}. This also holds for $k = 1$, where $L_{i+1} = L_i$. □

3. The persistence order

Let $k \geq 1$, let Λ be a finite set of labels, and let $A_i \subseteq \Lambda$ for $i \in \mathbb{Z}_k$; we think of A_i as the set of labels active in row i. For $\lambda \in A_i$ define the *persistence*

$$h_i(\lambda) = \begin{cases} \infty & \text{if } \lambda \in A_j \text{ for all } j \in \mathbb{Z}_k, \\ \max\{s \geq 0 : \ \lambda \in A_{i+1} \cap A_{i+2} \cap \cdots \cap A_{i+s}\} & \text{otherwise.} \end{cases}$$

For $s = 0$ the condition in the second case is empty. In the second case $0 \leq h_i(\lambda) \leq k-2$, since $\lambda \in A_i$ and some row does not contain λ. Fix a total order $\prec$ on Λ. The *persistence order* of A_i lists its elements by decreasing h_i, with ties broken by $\prec$: thus λ precedes μ if $h_i(\lambda) > h_i(\mu)$, or $h_i(\lambda) = h_i(\mu)$ and $\lambda \prec \mu$. Let $\mathrm{pos}_i(\lambda)$ denote the position of λ in the persistence order of A_i.

Lemma 3.1. *If $\lambda \in A_i \cap A_{i+1}$, then $\mathrm{pos}_i(\lambda) \leq \mathrm{pos}_{i+1}(\lambda)$.*

Proof. For $\mu \in A_i \cap A_{i+1}$ we have $h_{i+1}(\mu) = \infty$ if $h_i(\mu) = \infty$, and $h_{i+1}(\mu) = h_i(\mu) - 1$ otherwise. Indeed, in the second case $s = h_i(\mu)$ satisfies $1 \leq s \leq k-2$, the label μ lies in $A_{i+1}, \ldots, A_{i+s}$ and not in A_{i+s+1}, and these rows are distinct; hence $h_{i+1}(\mu) = s - 1$.

Let μ precede λ in the persistence order of A_i. Since $\lambda \in A_{i+1}$, we have $h_i(\lambda) \geq 1$, whereas every $\mu \in A_i \setminus A_{i+1}$ has $h_i(\mu) = 0$ and therefore follows λ. Hence $\mu \in A_{i+1}$. By the first paragraph, passing from h_i to h_{i+1} lowers all finite values among λ, μ by one and keeps the value ∞; this preserves strict inequalities and equalities between $h(\mu)$ and $h(\lambda)$. Since the tie-breaking order is the same in both rows, μ precedes λ in the persistence order of A_{i+1} as well. Thus every predecessor of λ in row i is a predecessor of λ in row $i+1$, which proves the lemma. □

The proof uses neither equal sizes of the sets A_i nor any condition on labels entering at row $i+1$; such labels can only add predecessors.

Lemma 3.2. *Let $\lambda_1 \prec \lambda_2 \prec \cdots \prec \lambda_m$ be labels such that, for every $i \in \mathbb{Z}_k$ and $1 \leq j < m$, $\lambda_{j+1} \in A_i$ implies $\lambda_j \in A_i$. Then in the persistence order of every A_i, the labels λ_j lying in A_i appear in increasing order of j.*

Proof. Let $j < j'$ with $\lambda_j, \lambda_{j'} \in A_i$. Every row containing $\lambda_{j'}$ contains λ_j, so $h_i(\lambda_j) \geq h_i(\lambda_{j'})$. Ties are broken by $\prec$, and $\lambda_j \prec \lambda_{j'}$. □

4. Proof of the main theorem

Proposition 4.1. *Let v be a node of a capacity tree, let $k \geq 1$, and let $I_1, \ldots, I_k$ be pairwise disjoint independent sets of N_v. Then there are linear orders L_i of I_i such that, with $c = c_v$, every window $W_{i,t}$, $i \in \mathbb{Z}_k$, $0 \leq t \leq c$, is independent in N_v.*

Every member of $\mathcal{I}_v$ has at most c_v elements, so the windows are defined.

Proof. We use induction on the height of v in the tree. If v is a leaf, take any orders. Every window has at most c elements by Lemma 2.4, hence is independent in N_v.

Let v be an internal node, and suppose that the proposition holds for every child of v. For a child u of v and $i \in \mathbb{Z}_k$ put $I_i^u = I_i \cap E_u$. By Definition 2.1, I_i^u is independent in N_u, and these sets are pairwise disjoint for fixed u. By the induction hypothesis there are linear orders L_i^u of I_i^u whose windows

$$W_{i,s}^u = (L_i^u)^{>s} \cup (L_{i+1}^u)^{\leq s}, \qquad 0 \leq s \leq c_u,$$

are independent in N_u. Let $m_{i,u} = |I_i^u|$; note that $m_{i,u} \le c_u$. Write $x_{i,u,1}, \dots, x_{i,u,m_{i,u}}$ for the elements of I_i^u in the order L_i^u.

The merged order. Let Λ be the set of pairs (u, j) with u a child of v and $j \ge 1$ an integer with $j \le m_{i,u}$ for some i. Let

$$A_i = \{(u, j) \in \Lambda : j \le m_{i,u}\} \qquad (i \in \mathbb{Z}_k).$$

Since the sets E_u partition E_v, the map $(u, j) \mapsto x_{i,u,j}$ is a bijection from A_i onto I_i. Fix an order of the children of v and let $\prec$ be the induced lexicographic order on Λ, so that $(u, j) \prec (u, j')$ whenever $j < j'$. Let L_i be the persistence order of A_i with every label (u, j) replaced by $x_{i,u,j}$.

Step 1: the merge preserves the orders of the children. For each child u, the labels $(u, 1) \prec (u, 2) \prec \cdots$ satisfy the hypothesis of Lemma 3.2, because $(u, j+1) \in A_i$ means $j + 1 \le m_{i,u}$. Hence the restriction of L_i to E_u is L_i^u.

Step 2: reduction of a window to the children. Fix $i \in \mathbb{Z}_k$ and $0 \le t \le c$, and fix a child u. Let $m = m_{i,u}$, let a be the number of elements of E_u in $L_i^{\le t}$, and let b be the number of elements of E_u in $L_{i+1}^{\le t}$. Since $L_i^{\le t}$ is an initial segment of L_i, Step 1 shows that $L_i^{\le t} \cap E_u = (L_i^u)^{\le a}$, and similarly for row $i + 1$. Therefore

$$W_{i,t} \cap E_u = (L_i^u)^{>a} \cup (L_{i+1}^u)^{\le b}. \tag{4.1}$$

We claim that

$$a \ge \min(b, m). \tag{4.2}$$

This is clear if $b = 0$ or $m = 0$. Suppose $1 \le b \le m$. The label $\lambda = (u, b)$ lies in A_i, since $b \le m$, and in A_{i+1}, since $b \le m_{i+1,u}$. The element $x_{i+1,u,b}$ of row $i + 1$ carrying this label lies in $L_{i+1}^{\le t}$, so $\mathrm{pos}_{i+1}(\lambda) \le t$. By Lemma 3.1, $\mathrm{pos}_i(\lambda) \le t$, that is, $x_{i,u,b} \in L_i^{\le t}$. Hence $(L_i^u)^{\le a}$ contains $x_{i,u,b}$, and $a \ge b$. Suppose finally $b > m \ge 1$. Then $\lambda = (u, m)$ lies in $A_i \cap A_{i+1}$, and $x_{i+1,u,m}$ precedes $x_{i+1,u,b}$ in L_{i+1} by Step 1; hence it lies in $L_{i+1}^{\le t}$, and $\mathrm{pos}_{i+1}(\lambda) \le t$. As before $x_{i,u,m} \in L_i^{\le t}$, so $a \ge m$. This proves (4.2).

If $a = m$, then $(L_i^u)^{>a} = \emptyset$, and (4.1) is a subset of I_{i+1}^u, which is independent in N_u. If $a < m$, then (4.2) gives $b \le a$, and $a < m \le c_u$. Hence $W_{i,a}^u$ is one of the windows provided by the induction hypothesis, and $(L_{i+1}^u)^{\le b} \subseteq (L_{i+1}^u)^{\le a}$ shows that

$$W_{i,t} \cap E_u \subseteq (L_i^u)^{>a} \cup (L_{i+1}^u)^{\le a} = W_{i,a}^u.$$

So $W_{i,t} \cap E_u$ is independent in N_u in both cases.

Step 3: conclusion. By Step 2, $W_{i,t} \cap E_u$ is independent in N_u for every child u, and $|W_{i,t}| \le c$ by Lemma 2.4. By Definition 2.1, $W_{i,t}$ is independent in N_v. □

Proof of Theorem 1.1. If $r = 0$, then $E = \emptyset$ and there is nothing to prove, so let $r > 0$. Take a capacity tree for M as in Lemma 2.2, so that the root has capacity r and $N_{\mathrm{root}} = M$. The bases $B_1, \dots, B_k$ are pairwise disjoint independent sets of N_{root}, and Proposition 4.1 gives linear orders L_i of B_i whose windows $W_{i,t}$, $0 \le t \le r$, are independent in M. Since $|B_i| = r$ for all i, Lemma 2.4 shows that every set of r cyclically consecutive entries of $L_1 L_2 \cdots L_k$ is one of these windows. Its entries are r distinct elements, because the B_i are disjoint, so it is an independent set of size r, that is, a basis. □

The proof does not use that the bases cover E. We record this for later use.

Corollary 4.2. *Let M be a laminar matroid of rank r and let $B_1, \dots, B_k$ be pairwise disjoint bases of M. Then there are linear orders of the B_i such that every r cyclically consecutive entries of the concatenation $B_1 B_2 \cdots B_k$ form a basis.*

Remark 4.3. The proof is constructive. Given a laminar presentation $(E, \mathcal{A}, c)$ and the bases, the capacity tree of Lemma 2.2 is obtained directly from the inclusion relation on $\mathcal{A}$. The orders are then computed from the leaves to the root: at an internal node one computes the persistence of each label, which depends only on the lengths $m_{i,u}$ of the rows of the children, and sorts each row. Thus the orders are obtained in time polynomial in $|E|$, $|\mathcal{A}|$ and k, and the order of the blocks is never changed.

5. Relation to known results

5.1. **Two bases.** A matroid is *strongly base orderable* if for any two bases A and B there is a bijection $\varphi\colon A \to B$ such that $(A \setminus X) \cup \varphi(X)$ is a basis for every $X \subseteq A$. Applying this to $A \setminus X$ shows that $(B \setminus \varphi(X)) \cup X$ is a basis as well. If A and B are disjoint, order A as $a_1, \dots, a_r$ and B as $\varphi(a_1), \dots, \varphi(a_r)$. With $X = \{a_1, \dots, a_t\}$, the windows of Definition 2.3 are $W_{1,t} = (A \setminus X) \cup \varphi(X)$ and $W_{2,t} = (B \setminus \varphi(X)) \cup X$, so every window is a basis; compare [1, Section 1]. As laminar matroids are strongly base orderable, Theorem 1.1 for $k \le 2$ is not new. For $k \ge 3$ one may choose such bijections between consecutive blocks, but the resulting orders need not agree after a full turn around the cycle.

5.2. **Transversal matroids.** Let M be a transversal matroid of rank r. Then M has a presentation by r sets $A_1, \dots, A_r$, see [10, Section 1.6] and [2, Section 3.1], and every basis is a transversal of $(A_1, \dots, A_r)$. Given disjoint bases $B_1, \dots, B_k$, choose for each i a bijection $\sigma_i\colon B_i \to \{1, \dots, r\}$ with $x \in A_{\sigma_i(x)}$, and list B_i in the order $\sigma_i^{-1}(1), \dots, \sigma_i^{-1}(r)$. Every set of r cyclically consecutive entries then consists of elements assigned to the slots $t+1, \dots, r$ in one block and $1, \dots, t$ in the next, hence is a transversal. This elementary argument proves the analogue of Theorem 1.1 for transversal matroids and any k. It does not apply to laminar matroids, which need not be transversal (Proposition 5.2).

5.3. **A separating example.** Let P_1, P_2, P_3, P_4 be pairwise disjoint sets of size three, let $E = P_1 \cup \dots \cup P_4$, and let M be the matroid on E whose independent sets are the sets X with $|X \cap P_j| \le 1$ for all j and $|X| \le 3$. This is the case $k = 4$ of the matroid used by Bérczi, Schwarcz and Yamaguchi to show that their bound for gammoids is tight [2, proof of Theorem 1.10]. Write $P_j = \{p_{j,1}, p_{j,2}, p_{j,3}\}$ and, for $i \in \{1, 2, 3, 4\}$, let

$$B_i = \{p_{j,s} :\ j \ne i,\ s \equiv i - j \pmod 4,\ 1 \le s \le 3\}.$$

Then B_i contains one element of each P_j with $j \ne i$, and $p_{j,s}$ lies only in B_{j+s}, indices modulo 4. So $B_1, \dots, B_4$ are bases partitioning E.

We recall the definition of split matroids from [1, Section 2]. Let S be a finite set, let $H_1, \dots, H_q \subseteq S$, and let $r, r_1, \dots, r_q$ be nonnegative integers with $|S| \ge r$,

$$|H_i \cap H_j| \le r_i + r_j - r \quad (i \ne j), \qquad |S \setminus H_i| + r_i \ge r \quad (1 \le i \le q). \tag{5.1}$$

The *elementary split matroid* defined by these data has as bases the r-element sets $X \subseteq S$ with $|X \cap H_i| \le r_i$ for all i. A *split matroid* is the direct sum of an elementary split matroid and uniform matroids.

Lemma 5.1. *A set $X \subseteq S$ is independent in the elementary split matroid defined above if and only if $|X| \le r$ and $|X \cap H_i| \le r_i$ for all i.*

Proof. Subsets of bases satisfy both conditions. Conversely, let X satisfy them and $|X| < r$. If X were tight for two distinct indices, that is, $|X \cap H_i| = r_i$ and $|X \cap H_j| = r_j$, then by (5.1)

$$|X| \ge |X \cap (H_i \cup H_j)| \ge r_i + r_j - |H_i \cap H_j| \ge r,$$

a contradiction. If X is tight for no index, adding any element of $S \setminus X$, which is nonempty since $|S| \geq r > |X|$, preserves all constraints. If X is tight for exactly one index i, then some element of $S \setminus H_i$ lies outside X, since otherwise $|X| \geq |S \setminus H_i| + r_i \geq r$; adding it preserves all constraints. Repeating this, we extend X to a basis. □

Proposition 5.2. *The matroid M is a connected laminar matroid of rank three. It is the cycle matroid of the graph obtained from a cycle of length four by replacing every edge with three parallel edges. It is neither transversal nor split.*

Proof. $M = M(E, \{P_1, \ldots, P_4, E\}, c)$ with $c(P_j) = 1$ and $c(E) = 3$, and it has rank three. Mapping the elements of P_j to the three parallel copies of the j-th edge of the 4-cycle identifies the forests of the graph with the independent sets of M. Two elements of the same class form a circuit, and elements of distinct classes P_i, P_j lie in the circuit consisting of one element from each class. So any two elements lie in a common circuit, and M is connected.

Suppose that M is transversal, presented by a bipartite graph between E and a set T, where $X \subseteq E$ is independent if and only if it can be matched into T. Every element has a nonempty neighbourhood $\Gamma(x)$, since M has no loops. If x, y are parallel, then $\{x, y\}$ cannot be matched, so $|\Gamma(x) \cup \Gamma(y)| \leq 1$ and $\Gamma(x) = \Gamma(y)$ is a single vertex. Hence all elements of P_j have the same neighbourhood $\{\tau_j\}$. If $\tau_i = \tau_j$ for some $i \neq j$, then a pair consisting of an element of P_i and one of P_j could not be matched, although it is independent. So $\tau_1, \ldots, \tau_4$ are distinct, and one element from each class can be matched into T. This independent set has four elements, contradicting $r(M) = 3$.

Suppose that M is split. Since M is connected and not uniform, it is an elementary split matroid; see also [1, Lemma 6]. Let it be defined by data as in (5.1) with $r = 3$. Take two elements $x, y \in P_1$. By Lemma 5.1, the dependent set $\{x, y\}$ violates a constraint, so there is an index i with $|\{x, y\} \cap H_i| > r_i$. If $r_i = 0$, an element of $\{x, y\} \cap H_i$ would be a loop; hence $r_i = 1$ and $x, y \in H_i$. Similarly there is an index j with $r_j = 1$ and H_j containing two elements $x', y' \in P_2$. If $i = j$, then the independent set $\{x, x'\}$ would violate $|\{x, x'\} \cap H_i| \leq 1$; so $i \neq j$. Now (5.1) gives $|H_i \cap H_j| \leq 1 + 1 - 3 < 0$, which is impossible. □

Example 5.3. For the partition $B_1, \ldots, B_4$ above, the capacity tree of Lemma 2.2 has the root E of capacity three and the leaves $P_1, \ldots, P_4$ of capacity one. The label of the unique element of $B_i \cap P_j$ is $(P_j, 1)$, active in the three rows $i \neq j$, and its persistence in row i is the residue of $j - i - 1$ modulo 4. Hence the persistence order lists $B_i \cap P_{i-1}, B_i \cap P_{i-2}, B_i \cap P_{i-3}$, indices modulo 4, and the construction yields the cyclic sequence whose classes are

$$4, 3, 2 \;\mid\; 1, 4, 3 \;\mid\; 2, 1, 4 \;\mid\; 3, 2, 1,$$

where the bars separate $B_1, \ldots, B_4$. Every three cyclically consecutive entries lie in distinct classes, and so form a basis.

A cyclic ordering of E alone is easy to obtain for this matroid, for instance as in Example 5.3. It also follows from a theorem of Liang, Locke and Trice [9, Theorem 3.2(i)] on graphs, applied with $i = 3$: the graph of Proposition 5.2 has twelve edges, at most three parallel edges between two vertices, and its simplification is a cycle of length four. The point of Theorem 1.1 is that an arbitrary given partition into bases and an arbitrary order of its blocks are kept. Proposition 5.2 shows that, already in rank three, laminar matroids lie outside the scope of both the transversal argument of Section 5.2 and the theorem of [1] for split matroids.

5.4. **Further remarks.** When $|E| = kr$ with $r > 1$, the integers $|E|$ and r are not coprime. In this case the theorem of van den Heuvel and Thomassé [7], in the form [1, Theorem 14], provides a partition of E into sets of size $\gcd(|E|, r) = r$, that is, a partition into bases, and

gives no information on the order inside the blocks. Fletcher [5, Theorem 1.2 and Corollary 1.3] studies cyclic orderings of uniformly dense matroids of rank three in which no basis partition is prescribed; the difference from the prescribed setting is noted in [5, Remark 2.5].

Remark 5.4. Following [1, Section 4.2], a *cyclic exchange* in a sequence of bases replaces bases $B_{i_1}, \dots, B_{i_q}$ with distinct indices by $B_{i_j} - e_j + e_{j+1}$, where $e_j \in B_{i_j}$, the indices j are taken modulo q, and all new sets are bases. Let $k \geq 2$, let $B_1, \dots, B_k$ be pairwise disjoint bases of a laminar matroid of rank r, ordered as in Corollary 4.2, and let $e_i(s)$ be the s-th element of B_i. For $0 \leq t \leq r$ the windows $W_{1,t}, \dots, W_{k,t}$ are bases, and

$$W_{i,t+1} = W_{i,t} - e_i(t+1) + e_{i+1}(t+1) \qquad (0 \leq t < r).$$

Hence the sequence $(B_1, \dots, B_k) = (W_{1,0}, \dots, W_{k,0})$ is transformed into $(B_2, \dots, B_k, B_1) = (W_{1,r}, \dots, W_{k,r})$ by r cyclic exchanges. This number is optimal, as noted at the end of [1, Section 4.2]: a cyclic exchange changes each basis of the sequence by at most one element, whereas the first basis has to be changed from B_1 into the disjoint set B_2 of size r. We do not consider the general question of White [11] on transforming one sequence of bases into a compatible one.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

References

[1] K. Bérczi, Á. Jánosik, and B. Mátravölgyi, *Cyclic ordering of split matroids*, Electron. J. Combin. **33** (2026), no. 1, Paper No. P1.20.

[2] K. Bérczi, T. Schwarcz, and Y. Yamaguchi, *List coloring of two matroids through reduction to partition matroids*, SIAM J. Discrete Math. **35** (2021), no. 3, 2192–2209.

[3] T. Fife and J. Oxley, *Laminar matroids*, European J. Combin. **62** (2017), 206–216.

[4] L. Finkelstein, *Two algorithms for the matroid secretary problem*, Master's thesis, Technion, Israel Institute of Technology, Haifa, 2011.

[5] A. Fletcher, *Cyclic basis orderings of uniformly dense rank-three matroids*, version-2 draft dated 10 September 2026, deposited on Zenodo, doi:10.5281/zenodo.22698709.

[6] H. Gabow, *Decomposing symmetric exchanges in matroid bases*, Math. Programming **10** (1976), no. 1, 271–276.

[7] J. van den Heuvel and S. Thomassé, *Cyclic orderings and cyclic arboricity of matroids*, J. Combin. Theory Ser. B **102** (2012), no. 3, 638–646.

[8] Y. Kajitani, S. Ueno, and H. Miyano, *Ordering of the elements of a matroid such that its consecutive w elements are independent*, Discrete Math. **72** (1988), no. 1–3, 187–194.

[9] Y. Liang, S. Locke, and A. Trice, *Cyclic ordering of edges without long acyclic subsequences*, Congr. Numer. **235** (2025), 41–46.

[10] J. Oxley, *Matroid theory*, second ed., Oxford Graduate Texts in Mathematics, vol. 21, Oxford University Press, Oxford, 2011.

[11] N. L. White, *A unique exchange property for bases*, Linear Algebra Appl. **31** (1980), 81–91.

[12] D. Wiedemann, *Cyclic base orders of matroids*, manuscript dated 9 May 2006, typewritten version of a handwritten note of August 1984, archived at `https://web.archive.org/web/20121004065333id_/http://www.plumbyte.com:80/cyclic_base_orders_1984.pdf`.

Knowledge Sense
Email address: `math@knowledgesense.jp`